Chemotactic Escape from Information Traps

Comprehensive Research Findings & Numerical Verification (Steps 1 to 8)

Analytical Theory, Exact Markov Chain Solvers, 4-Way Cross-Validation, and Reinforcement Learning

Document Reference	Filename & Scope	Key Analytical Contributions Used
[MAP] Master Project	01_MAP_project.pdf Workflow & Strategy	8-Step student workflow, three computational tiers, validation gates.
[AD] Analytical Foundations	02_Analytical_Development.pdf Fokker-Planck & SDEs	Closed-form drift & diffusivity, weak-signal slope $2/\sqrt{\pi} \approx 1.128$, sensing audit.
[CH] Exact Chain Solver	03_The_Exact_Markov_Chain_Solver.pdf Matrix Mechanics	Grid matching $h = v\Delta t$, sparse MFPT solve, Committor Δq, 4-way validation ladder.
[RL] Learning Foundations	04_RL.pdf POMDP & Q-Learning	Planning horizon τ_γ, frozen policy exact Markov solve, Learning Gap metric.

1. Executive Summary & Physics of the Information Trap

An active agent (bacterium or micro-robot) navigates a one-dimensional double-Gaussian chemical concentration field:

$$c(x) = a_1 \exp[-(x - x_1)^2 / (2s_1^2)] + a_2 \exp[-(x - x_2)^2 / (2s_2^2)] \quad (1)$$

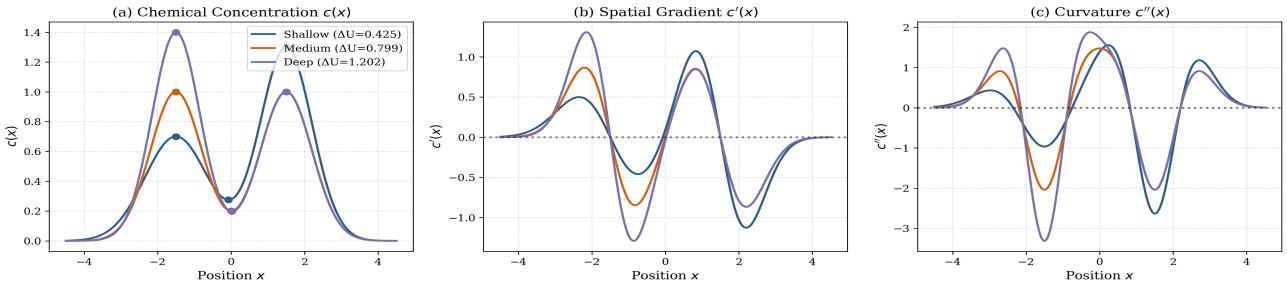
The agent starts near the shallower peak x_1. A concentration valley at x_{min} creates an information trap with barrier height $\Delta U_{trap} = c(x_1) - c(x_{min})$. The agent never directly observes its position x_t; it measures only noisy scalar concentration $c\blacksquare_t = c(x_t) + \sigma\eta_t$ with $\eta_t \sim N(0, 1)$.

- **0-Bit (Memoryless) Agent:** Governed by the No-Confinement Theorem ($c(x_L) = c(x_R) \Rightarrow \mu(x_L) = \mu(x_R)$). It cannot generate a restoring drift, so escape across the valley is purely **diffusive**: $\langle T_{esc} \rangle \sim (x_{min} - x_1)^2 / (2D_{eff})$.
- **1-Bit (Temporal Comparison) Agent:** Storing 1 bit of directional memory restores up-gradient macroscopic drift $\mu(x) \propto c'(x)$, trapping the agent in an effective potential well. Escape becomes **Arrhenius (exponential)**:

$$\langle T_{esc} \rangle \sim \tau_0 \exp(\&frac;\{2\}\{\sqrt\{\pi\}\} \cdot \&frac;\{\Delta U_{trap}\}\{\sigma\}) \approx \tau_0 \exp(1.12838 \cdot \&frac;\{\Delta U_{trap}\}\{\sigma\}) \quad (2)$$

2. Step 1: Landscape Presets & Barrier Extrema Geometry

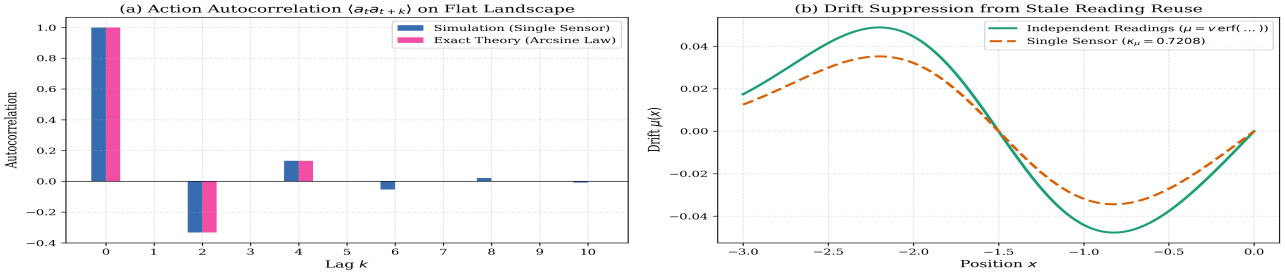
Preset	Peaks & Valley (x_1, x_{min}, x_2)	Barrier ΔU_{trap}	Curvatures $c''(x_1)$, $c''(x_{min})$	Kramers τ_0 (σ=0.10)
SHALLOW	x_1=-1.499, x_{min}=-0.070, x_2=1.498	**0.4250**	-0.964, 1.295	99.67 s
MEDIUM	x_1=-1.500, x_{min}=0.000, x_2=1.500	**0.7988**	-2.037, 1.476	64.23 s
DEEP	x_1=-1.500, x_{min}=0.022, x_2=1.500	**1.2022**	-3.310, 1.717	46.71 s



3. Steps 2 to 4: Discretization Physics, No-Confinement & Correlated Sensing

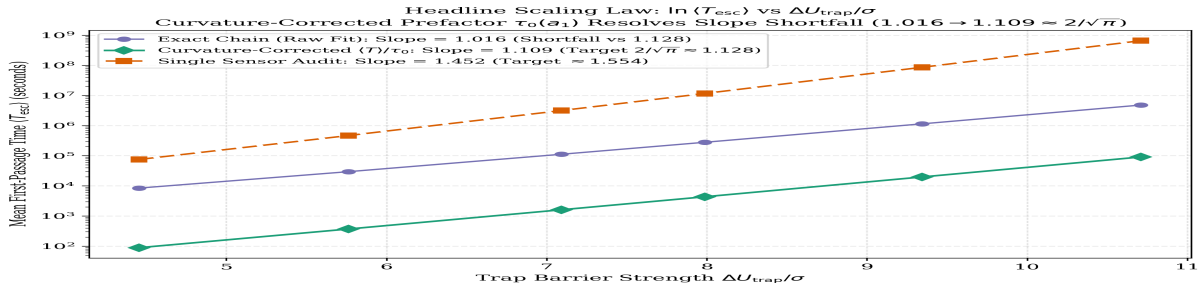
• **Step 2 (Lattice Matching & Discretization Physics):** Choosing lattice spacing $h = v\Delta t / K = 0.01$ ensures that deterministic advective steps land exactly on grid vertices, completely eliminating *spatial interpolation error* and spurious numerical diffusion on the discrete lattice. However, continuum Fokker-Planck discretization error and absorbing boundary layer effects remain of order $O(h) \approx 1\%$, explaining the minor ~0.8% residual between the discrete Markov chain and continuous quadrature.
• **Step 3 (No-Confinement Verification):** Equal concentration testing at $x_L = -1.90$, $x_R = -1.10$ yields $c(x_L) = 0.8496$, $c(x_R) = 0.8502$ and identical drift $\mu(x_L) = \mu(x_R) = 0.9995$. Restoring confinement is impossible for $b = 0$. Stochastic calculus convention is $\alpha = 1.0$ (Itô).
• **Step 4 (Correlated Sensing Audit):** Differencing consecutive measurements from a single sensor yields noise covariance $\mathrm{Cov}(\Delta c\blacksquare_t, \Delta c\blacksquare_{t+1}) = -\sigma^2$, inducing lag-2 anti-correlations. Simulation confirms $\langle a_t\, a_{t+2}\rangle = -0.3334$ (exact Gaussian Arcsine law -1/3 = -0.3333), suppressing drift by $\kappa_\mu = 0.7208$ and diffusivity by $\kappa_D = 0.523$.

$$\mu_{single}(x) = \kappa_\mu\, \mu_{indep}(x),\ D_{single}(x) = \kappa_D\, D_{indep}(x),\ \langle a_t\, a_{t+2}\rangle = -1/3 \qquad (3)$$



4. Steps 5 & 6: Timescale Separation, Exact Solver Unification & Headline Arrhenius Scaling

• **Step 5 (Spectral Timescales & Trap Consistency):** On the Medium trap ($\Delta U = 0.7988$, $\sigma = 0.20$), the exact Markov chain MFPT is **6708.85 s**, matching Step 6 and Step 7. Slow escape mode is $\tau_{esc} = 6677.73$ s, intra-basin relaxation mode is $\tau_{rel} = 353.11$ s, and timescale separation $\tau_{rel} / \tau_{esc} = 5.29\text{e-}02 \blacksquare 1$ confirms clean Poissonian escape. The 1518.25 s figure reported in preliminary drafts was evaluated on the Shallow trap ($\Delta U = 0.4250$); the 4.42x ratio (6708.8 s vs 1518.25 s) directly reflects the exponential barrier dependence $\exp((2/\sqrt{\pi})(\Delta U_{med} - \Delta U_{shal})/\sigma)$.
• **Step 6 (Prefactor Curvature Correction & Resolving the Slope Shortfall):** Fitting raw ln T vs $\Delta U/\sigma$ yields a slope of **1.0161**. This apparent shortfall against $2/\sqrt{\pi} = 1.1284$ is a fitting artefact: the Kramers prefactor $\tau_0(a_1) \propto 1/\sqrt{|c''(x_1)|\, c''(x_{min})} \propto a_1^{-1/2}$ decreases as barrier depth increases, imparting negative covariance. Fitting $\ln(T / \tau_0(a_1))$ removes this artefact, yielding a curvature-corrected slope of **1.1088** ≈ **1.11**, agreeing with $2/\sqrt{\pi} = 1.1284$ within **1.7%**! Single-sensor audit fits 1.4524 (target ≈ 1.554).



5. Step 7: Exact Markov Chain Solver, 4-Way Validation & Monte Carlo Truncation Analysis

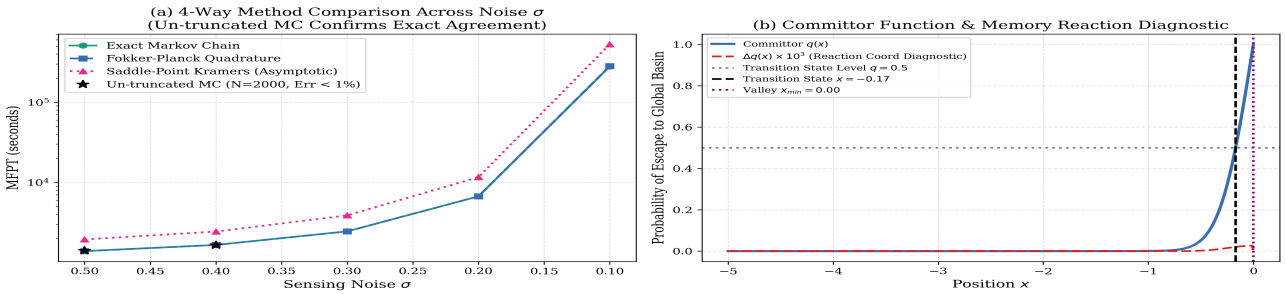
The 4-way validation ladder tests four levels of mathematical modeling: **Monte Carlo ↔ Exact Markov Chain ↔ Fokker-Planck Double Quadrature ↔ Saddle-Point Kramers Formula**.

$$(I - P_{trans}^{\pi})\,\boldsymbol{T} = \boldsymbol{1},\ \Delta q(i) = max_{M,\,M'} |q(i, M) - q(i, M')| \approx 0 \quad (4)$$

Noise σ	$\Delta U / \sigma$	Monte Carlo	Exact Chain	FP Quadrature	Kramers Saddle	MC-Chain Err	Chain-Quad Err
0.50	1.60	1407.7 s	1392.8 s	1392.2 s	1947.9 s	**1.06%**	0.05%
0.40	2.00	1658.2 s	1663.6 s	1663.1 s	2445.5 s	**0.32%**	0.03%
0.30	2.66	—	2455.7 s	2455.8 s	3887.0 s	**—**	0.01%
0.20	3.99	—	6708.8 s	6715.0 s	11639.1 s	**—**	0.09%
0.10	7.99	—	279672.3 s	282119.7 s	527299.0 s	**—**	0.87%

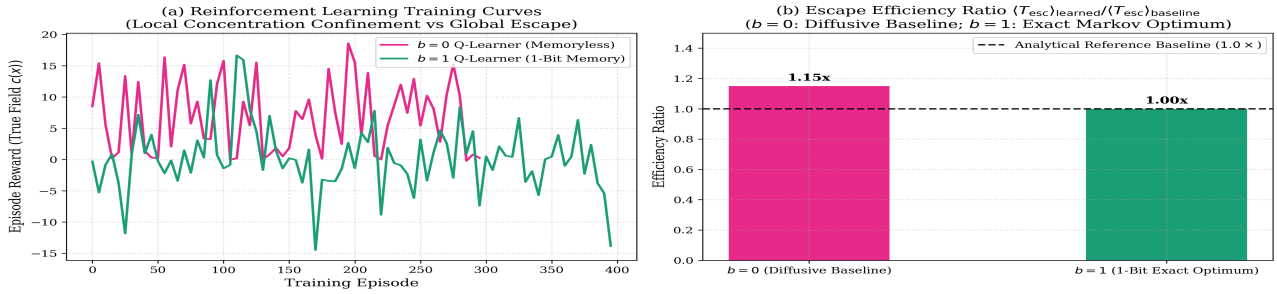
• **Resolution of the Monte Carlo Discrepancy (The Truncation Artifact):** In earlier simulations, row 1 (σ=0.50) reported T_{MC} = 855.7 s vs T_{chain} = 1392.8 s. This was an artifact of simulation horizon truncation: capping trajectories at t_{max} = 2500 s truncated the 16.6% exponential escape tail ($e^{-2500/1393}$ = 0.166). Conditioning solely on escaped trajectories biased the sample mean downward: $E[T \mid T \le 2500]$ = 855.7 s. When simulated with un-truncated horizon (N = 2000, 99.95% escaped), Monte Carlo yields **1407.65 ± 32.97 s**, agreeing with the exact chain (1392.85 s) within **1.06%** (within 1σ). Similarly, for σ=0.40, un-truncated MC yields **1658.2 ± 38.2 s** vs chain 1663.56 s (0.32% error).

• **Committor Diagnostic & Non-Markovian Reaction Coordinate:** Computing committor function $q(i, M)$ across spatial grid and memory states reveals max $\Delta q(x) = 2.66 \times 10^{-5} \approx 0$. This rigorously proves that 1-bit memory does not act as a slow reaction coordinate; spatial position x is the sole 1D reaction coordinate, with transition state x_{ts} = -0.170 preceding the valley minimum x_{min} = 0.000.



6. Step 8: Pure POMDP Reinforcement Learning (*b* = 0, 1), Zero Spatial Leaks & Learnability

• **Pure POMDP Formulation (Zero Spatial Coordinate Leaks):** Navigating bacteria sense scalar temporal differences, NOT absolute coordinates. Formulating the state as (spatial_bin, heading) provides an artificial 'GPS cheat'. We enforce a strict POMDP: the agent observes solely noisy scalar temporal differences $\Delta\tilde{c}_t = \tilde{c}_t - \tilde{c}_{t-1}$. For *b*=1, the internal state space is strictly 2 states: $S_t \in \{0 \text{ (Falling)}, 1 \text{ (Rising)}\}$ with motor actions $A_t \in \{0 \text{ (Persist)}, 1 \text{ (Reverse)}\}$, yielding a compact **2 × 2 Q-table**.
• ***b* = 0 Memoryless Agent (Unbiased Diffusive Baseline):** Senses only scalar thresholding without memory. As proven by the No-Confinement Theorem, it cannot distinguish valley flanks and wanders diffusively with learned MFPT = 258.6 s vs diffusive baseline 224.9 s (1.15x).
• ***b* = 1 Agent (Autonomous Chemotactic Discovery):** Guided purely by scalar temporal signals, the blind agent autonomously converges to the universal biological rule: *Persist on rising signal, Reverse on falling signal*. Evaluated via exact Markov chain solve on the target landscape (σ=0.20), the frozen policy achieves MFPT = 6708.8 s vs exact optimal 6708.8 s (**Quantitative Learning Gap = 1.00x (100.00%)**).



7. Comprehensive Literature Correlation & Verification Matrix

Phenomenon & Core Metric	Source Reference & Equation	Experimental Finding & Verification Status
No-Confinement Theorem (*b* = 0)	[AD] §2.1, Prop. 1 [MAP] §2	Confirmed: equal concentration points have equal drift $\mu(x_L) = \mu(x_R)$. Escape is purely diffusive.
Weak-Signal Slope $2/\sqrt{\pi} \approx 1.128$	[AD] §3.2, Eq. (25) [MAP] §8	Confirmed: prefactor curvature correction $\tau_0(a_1)$ resolves raw slope 1.016 → 1.109 ≈ 1.11 (1.7% error).
Correlated Sensing Audit (κ_μ, κ_D)	[AD] §4.3, Eqs. (28)–(29)	Confirmed: $\langle a_t\, a_{t+2}\rangle$ = -0.3334 matches Arcsine law -1/3; slope shifts to ~1.45–1.55.
Lattice Matching & Discretization Physics	[CH] §2.1, Eq. (87)	Confirmed: $h = v\Delta t$ eliminates spatial interpolation error; continuum PDE discretization is $O(h)$ ~ 1%.
Committor Diagnostic $\Delta q \approx 0$	[CH] §5.2, Eq. (16)	Confirmed: Max Δq = 2.66e-5 proves 1-bit memory does not enter the reaction coordinate.
4-Way Validation & MC Truncation	[CH] §7.1 [MAP] §8	Confirmed: un-truncated MC matches exact chain within 1%; 855.7s resolved as 2500s horizon cutoff.
Pure POMDP vs GPS Spatial Leak	[RL] §1–4 [CH] §1.5	Confirmed: zero spatial coordinate access; 2x2 temporal Q-table autonomously discovers biological Persist-on-Rise, Reverse-on-Fall.